VASTRAVIEW: SYSTEM ARCHITECTURE

Decoupled Edge Anthropometry & Remote GPU Diffusion Try-On Pipeline

1. PRESENTATION LAYER (React 19 + TypeScript + Vite - Port 5173)

Webcam / Photo Capture

HTML5 Camera Stream / Drag-and-Drop
Validation (256px - 12MB limit)

Interactive Landmark Canvas

33-Point MediaPipe Overlay
GrabCut Silhouette & Ratios

Comparison Try-On Studio

Interactive Split Slider (Before/After)
Lossless Pan / Zoom & PNG Export

HTTP / JSON

Skeletal Data

Result PNG

2. CORE API & INTELLIGENCE LAYER (FastAPI 0.162 / Python 3.12 - Port 8000)

API Gateway & Auth

Argon2id Password Hashing
PyJWT / HttpOnly Cookies
CORS & Pydantic Filters

Anthropometry Engine

Google MediaPipe Pose Full (CPU)
Height Ground Calibration
Ramanujan Ellipse Slices
(Shoulder, Chest, Waist, Hip)

Brand Sizing Engine

Weighted Midpoint Matching
23 Official Retail Charts
Tolerance Fit Feedback
(Ideal, Snug, Loose)

VTON Orchestrator

Async Job Queue (.runtime/)
Lossless WebP Packaging
Semantic Face/BG Compositor
Self-Healing Tunnel Bridge

PyMongo / SQL

Secure Tunnel HTTPS

3. LOCAL PERSISTENCE LAYER

MongoDB Community 8.2

Port 27018
- 23 Sourced Brand Charts
- User Anthropometric Scans

PostgreSQL 17 & Storage

Port 5433 & /data/processed
- Async Try-On Job States
- 57 Garments & Lossless PNGs

4. REMOTE COMPUTE CLUSTER (Google Colab GPU)

Encrypted Tunnel

Cloudflare / ngrok
Auto Token Discovery
Heartbeat Monitor

FASHN VTON 1.5 Worker

NVIDIA T4 / A100 GPU (FP16)
PyTorch & Diffusers (576x864)
Deep Garment Warping